Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

Egypt's Tourism Under Fire: Measuring the Impact of Crises on Visitor Arrivals (1995 to 2019)

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Introduction

tourism is one of the **biggest money makers** in egypt. every year millions of tourists come to see the pyramids, the temples, and the beaches. but whenever something bad happens in the country, like a revolution or a security incident, tourist numbers drop really fast. we all know this happens but we never really **measured it properly with data**.
so for this project we wanted to actually look at the numbers and find out which crisis caused the **biggest damage**, how long it takes to recover, and whether the neighbors like Morocco and Tunisia get hit the same way. we used **real data from the World Bank** that goes from 1995 all the way to 2019.

Research Questions

we set out to answer four main questions with this project:

- which crisis caused the **biggest single year drop** in tourist arrivals?
- how long does it take for the numbers to recover after each crisis?
- do neighboring countries like **Morocco, Tunisia, and Jordan** get hit the same way?
- are the drops during crisis years **statistically significant** or just normal business changes?

Hypothesis

our main hypothesis was that the crisis years (like 2011, 2013, 2014, 2016) would show a **statistically significant difference** in tourism growth rates compared to stable years. in other words, we expected the drops to be **way too big** to be explained by normal ups and downs.
H0 (Null): there is no real difference in the average growth rate between stable years and turbulent years.
H1 (Alternative): there IS a real difference between the two groups.

Population of Interest

our population is **all international tourist arrivals to Egypt from 1995 to 2019**. this data comes from the **World Bank indicator ST.INT.ARVL** which tracks the total number of inbound tourists per year for every country. we also pulled data for Morocco, Tunisia, and Jordan as comparison countries.

Sampling Method

we did not use sampling in the traditional sense. we used the **entire population** of recorded tourist arrivals for egypt and three comparison countries. this is **census data** collected by national governments and reported to the World Bank. since we are using the full dataset and not a sample, there is **no sampling error** involved.

Bias Identification

since this is government reported data to the World Bank, there could be some **reporting bias**. some countries might count tourists differently or have gaps in certain years. we noticed a few **missing values** in the early years (before 1995) which is why we limited our analysis to 1995 onwards where the data was complete for all four countries.
another thing to keep in mind is that we picked the turbulent years (2011, 2013, 2014, 2016) based on known historical events. someone could argue that this is a form of **selection bias** since we already knew which years were bad. but we think this is fine because the whole point of the project is to measure the impact of specific known crises.

Survey Questions / Collected Data / Dataset

we used two CSV files from the World Bank:

- **arrival_data.csv**: contains annual tourist arrivals for 200+ countries from 1960 to 2024. each row is a country and each column is a year.
- **income_categories.csv**: contains metadata about each country including its region and income group.

Number of samples used: **25 years** of data per country (1995 to 2019) across **4 countries** (Egypt, Morocco, Tunisia, Jordan) = **100 data points** total.
the first thing we had to do was reshape the data. the world bank gives it in a wide format where every year is a separate column. we used **pd.melt()** to convert it into a **long format** where each row is one country in one year. then we calculated the **Year over Year percentage change** using **pct_change()** so we can compare drops across different decades fairly.

Analysis

the first visualization shows egypt's tourist arrivals from 1995 to 2019. you can clearly see the number growing from about 3 million in 1995 to a peak of **14.7 million in 2010**. after that

it **drops sharply in 2011** because of the arab spring, recovers a bit, then **drops again in 2016** after the sinai plane crash. by 2019 it was climbing back to about 13 million.

Figure 1: Egypt's tourism arrivals over time (1995 to 2019)

this next chart was actually surprising. most people would guess the arab spring in 2011 was the worst drop. but when you calculate the percentage change year over year, the sinai plane crash period (2016) was actually worse with a **42.1% drop**. the arab spring caused a **33.2% drop** and the 2008 financial crisis barely made a dent at just **2.3%**.

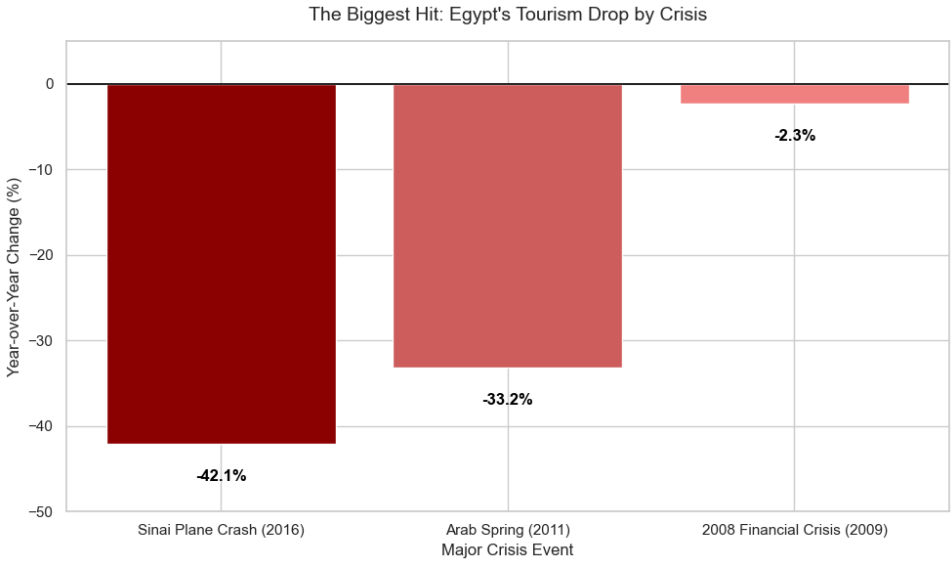

Figure 2: comparing the percentage drop for each major crisis

we also compared egypt with Morocco, Tunisia, and Jordan to see if the whole region suffers when egypt does. during the 2011 arab spring, both egypt and tunisia dropped sharply. but **morocco kept growing** the whole time like nothing happened. this tells us the crisis impact is **not automatic for the whole region**. it depends on what specifically happens in each country.

the **gray shaded area** in the chart marks the arab spring period (2011) so you can easily see who dropped and who didnt.

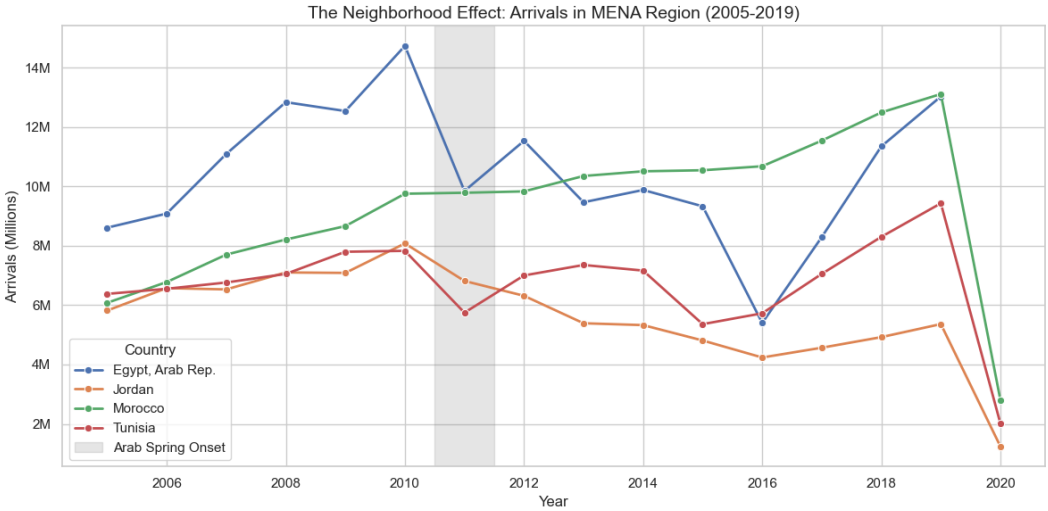

Figure 3: arrivals in egypt vs neighbors (2005 to 2019)

Hypothesis Testing Steps

Step 1: Define the hypotheses. **H0:** there is no real difference in the average YoY growth rate between stable years and turbulent years. **H1:** there IS a real difference between the two groups.

Step 2: Choose the test. we used an **independent two sample t test** (**ttest_ind** from scipy.stats). we chose the independent test instead of a paired one because the two groups have different numbers of years and they are separate time periods, not the same years measured twice.

Step 3: Calculate the results. **T statistic = 3.7622**, **P value = 0.0011**. the boxplot below shows how different the two groups are visually. stable years are mostly above zero (positive growth) while turbulent years are mostly deep in the negative.

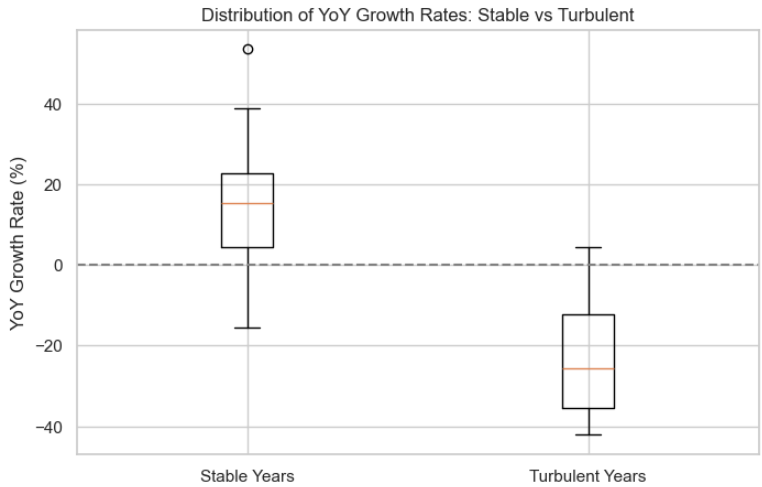

Figure 4: distribution of YoY growth rates, stable vs turbulent years

Step 4: Make a decision. since the p value (0.0011) is way less than our alpha of 0.05, we **reject the null hypothesis**. this means the drops during crisis years are NOT just random noise. they are mathematically proven to be different from normal years.

Conclusion

so what did we learn from all this? basically three things:

- the sinai plane crash period (2016) caused a bigger percentage drop (**42.1%**) than the arab spring (**33.2%**). thats not what most people would expect.
- **not all neighbors suffer the same way**. Morocco kept growing during 2011 while egypt and tunisia both dropped. so its not a region wide effect, it depends on what happens inside each specific country.
- the hypothesis test proved that crisis year drops are **statistically significant**. the p value was **0.0011** which is way below the 0.05 threshold. these are real shocks, not just normal business changes.

egypt's tourism does always come back eventually, which is good news. but the data shows that political and security events cause **serious damage** that takes years to fix.

Any Potential Issues

there are a few things we would like to point out about this analysis:

- the World Bank data only gives us total arrivals per year. it would be more useful if we had monthly data so we could see exactly when the drop starts after each crisis event.
- we picked the turbulent years manually based on historical events. a more rigorous approach would be to use an algorithm to detect outlier years automatically.
- the data stops at 2019 so it does not include COVID which was probably the biggest tourism disaster of all time. it would be interesting to extend this analysis to include 2020 onwards.

Extra Techniques and Sources

besides the stuff we learned in class, we also used a few extra techniques to make our charts cleaner and more professional. here are the youtube videos we used to learn them:

- **plt.axhline()** to draw baseline reference lines:
https://www.youtube.com/watch?v=aLXpwSqCSD8
- **sns.set_theme()** for global plot styling:
https://www.youtube.com/watch?v=6GUO8gZ0ZmM
- **f strings and enumerate()** for dynamic data labeling:
https://www.youtube.com/watch?v=Z8121lbP2Mc
- **plt.axvspan()** to shade crisis time periods on charts